Building a Pose Map

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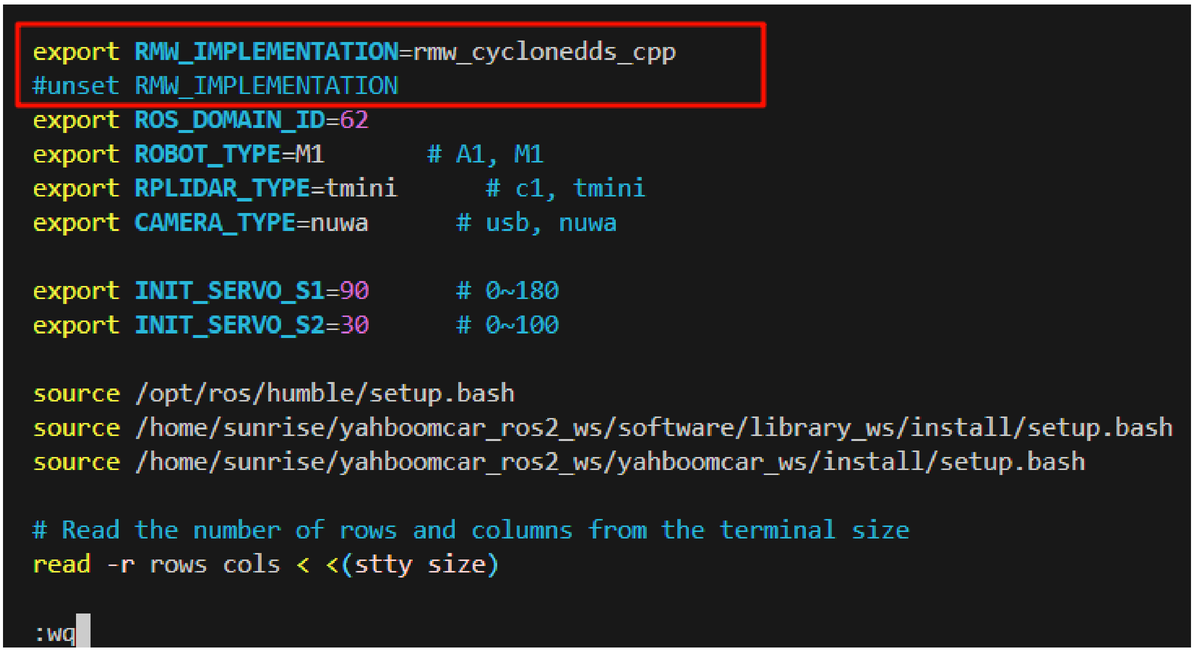
1. Building Raster Maps and Pose Maps using slam_toolbox

> [!NOTE]
>
> - At least one raster map is required before the road network .geojson file.
> - The pose map is used for fast relocalization during waypoint marking and navigation, improving the accuracy of global localization and thus the accuracy of waypoint marking.
> - This demonstration uses the **RDKX5 motherboard**. The rviz visualization is run in a virtual machine. The **ORIN motherboard** can run on the local machine.

2. Building a Raster Map

Open ~/.bashrc, uncomment this line, and change it as shown in the image. (After learning about road network planning, please restore this comment to avoid affecting previous examples.)

```
vim ~/.bashrc
:wq #Save and exit
source ~/.bashrc #Update environment
```



> [!TIP]
>
> - Before building the map, you can place some obstacles outside the roads on the map so that the actual position of the car can be referenced during navigation.

- Start slam_toolbox to build the map,

```
ros2 launch yahboomcar_nav map_slam_toolbox_launch.py
```

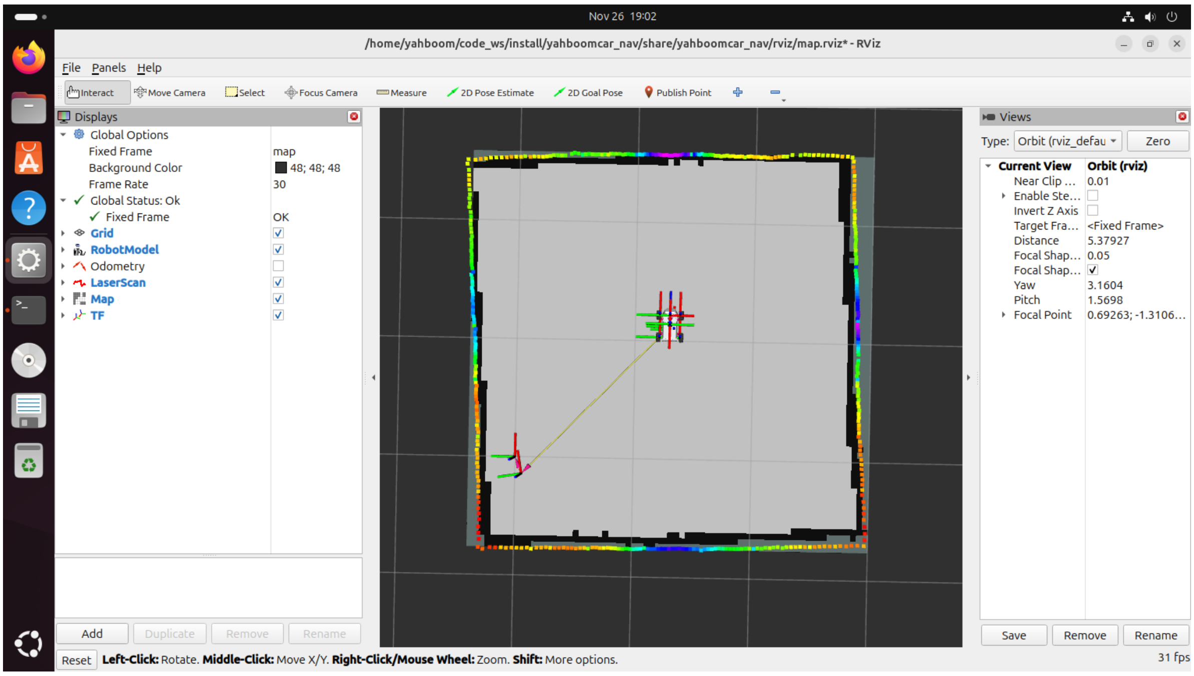
- Launch RViz visualization,

```
ros2 launch yahboomcar_nav display_map_launch.py
```

- Launch the keyboard control node (a gamepad can also be used),

```
ros2 run yahboomcar_ctrl yahboom_keyboard
```

- Control the car's movement to build the map

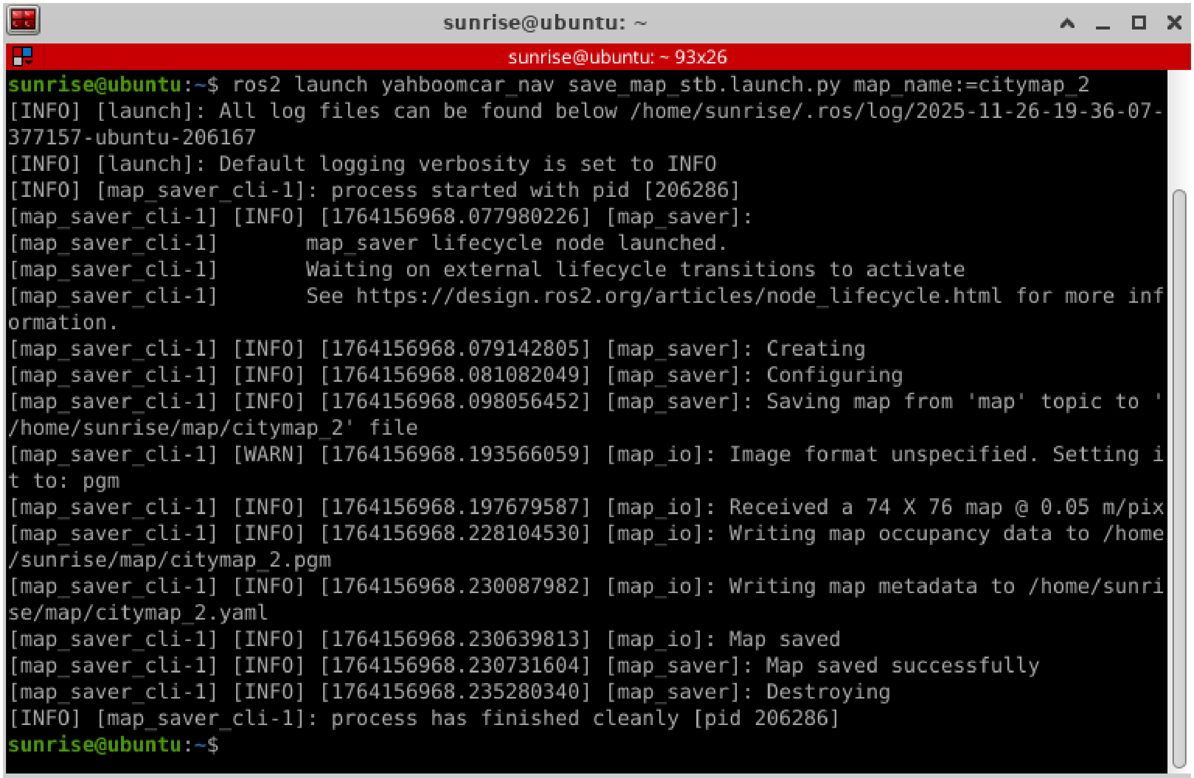


- Save the raster map

```
#RDKX5
ros2 launch yahboomcar_nav save_map_stb.launch.py map_name:=citymap_2
#ORIN
ros2 launch yahboomcar_nav save_map.launch.py map_name:=citymap_2
```

> [!TIP]
>
> - map_name specifies the **map name** for the saved raster map. If no startup parameter is specified, the default map name is `map`. All map files are saved in the `~/map` directory.

The terminal message `Map saved successfully` indicates that the map was saved successfully.



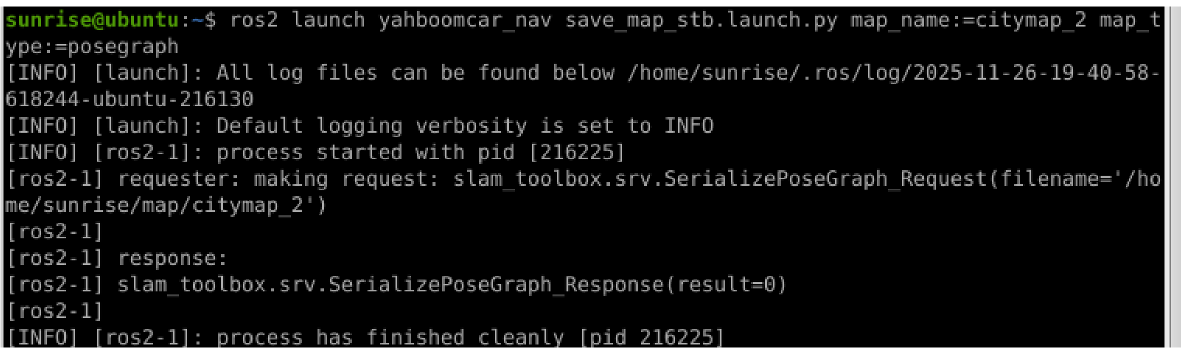
```
sunrise@ubuntu: ~
sunrise@ubuntu: ~ 93x26
sunrise@ubuntu:~$ ros2 launch yahboomcar_nav save_map_stb.launch.py map_name:=citymap_2
[INFO] [launch]: All log files can be found below /home/sunrise/.ros/log/2025-11-26-19-36-07-377157-ubuntu-206167
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [map_saver_cli-1]: process started with pid [206286]
[map_saver_cli-1] [INFO] [1764156968.077980226] [map_saver]:
[map_saver_cli-1]       map_saver lifecycle node launched.
[map_saver_cli-1]       Waiting on external lifecycle transitions to activate
[map_saver_cli-1]       See https://design.ros2.org/articles/node_lifecycle.html for more information.
[map_saver_cli-1] [INFO] [1764156968.079142805] [map_saver]: Creating
[map_saver_cli-1] [INFO] [1764156968.081082049] [map_saver]: Configuring
[map_saver_cli-1] [INFO] [1764156968.098056452] [map_saver]: Saving map from 'map' topic to '/home/sunrise/map/citymap_2' file
[map_saver_cli-1] [WARN] [1764156968.193566059] [map_io]: Image format unspecified. Setting it to: pgm
[map_saver_cli-1] [INFO] [1764156968.197679587] [map_io]: Received a 74 X 76 map @ 0.05 m/pix
[map_saver_cli-1] [INFO] [1764156968.228104530] [map_io]: Writing map occupancy data to /home/sunrise/map/citymap_2.pgm
[map_saver_cli-1] [INFO] [1764156968.230087982] [map_io]: Writing map metadata to /home/sunrise/map/citymap_2.yaml
[map_saver_cli-1] [INFO] [1764156968.230639813] [map_io]: Map saved
[map_saver_cli-1] [INFO] [1764156968.230731604] [map_saver]: Map saved successfully
[map_saver_cli-1] [INFO] [1764156968.235280340] [map_saver]: Destroying
[INFO] [map_saver_cli-1]: process has finished cleanly [pid 206286]
sunrise@ubuntu:~$
```

- Save the slam_toolbox pose map

```
#RDKX5
ros2 launch yahboomcar_nav save_map_stb.launch.py map_name:=citymap_2
map_type:=posegraph
#ORIN
ros2 launch yahboomcar_nav save_map.launch.py map_name:=citymap_2
map_type:=posegraph
```

> [!TIP]
>
> - `map_type` saves the map type. The default is `gridmap` for raster maps. Selecting `posegraph` saves the pose map of `slam_toolbox`.

The terminal displays `slam_toolbox.srv.SerializePoseGraph_Response(result=0)`, indicating the pose map has been successfully saved.



```
sunrise@ubuntu:~$ ros2 launch yahboomcar_nav save_map_stb.launch.py map_name:=citymap_2 map_type:=posegraph
[INFO] [launch]: All log files can be found below /home/sunrise/.ros/log/2025-11-26-19-40-58-618244-ubuntu-216130
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [ros2-1]: process started with pid [216225]
[ros2-1] requester: making request: slam_toolbox.srv.SerializePoseGraph_Request(filename='/home/sunrise/map/citymap_2')
[ros2-1]
[ros2-1] response:
[ros2-1] slam_toolbox.srv.SerializePoseGraph_Response(result=0)
[ros2-1]
[INFO] [ros2-1]: process has finished cleanly [pid 216225]
```

- Map files are saved by default in the `~/map` path. You can also specify the map name using the `map_name` parameter when saving the map.

```
sunrise@ubuntu:~$ ll ~/map/
total 7332
drwxrwxr-x  2 sunrise sunrise    4096 Nov 26 19:44 ./
drwxr-xr-x 42 sunrise sunrise    4096 Nov 26 18:24 ../
-rw-rw-r--  1 sunrise sunrise  275682 Nov 26 19:41 citymap_2.data
-rw-rw-r--  1 sunrise sunrise    5637 Nov 26 19:36 citymap_2.pgm
-rw-rw-r--  1 sunrise sunrise 7206000 Nov 26 19:41 citymap_2.posegraph
-rw-rw-r--  1 sunrise sunrise     129 Nov 26 19:36 citymap_2.yaml
```